Math 5601 - Independent Study Report

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Proposed Topic: *Adaptive Quadrature Methods for Numerical Integration*

1 Literature Review

Adaptive quadrature is a numerical integration technique that improves the accuracy of simple numerical methods by adaptively subdividing and bounding integration intervals. This method can effectively handle integrands that are not well-behaved, where traditional fixed-interval methods may fail. Composite integration inherently reduces the estimation error as the interval size is refined. Generally, if the error for an interval of size h is Ch^5, then halving the interval to $h/2$ reduces the error on each half to

$$C\left(\frac{h}{2}\right)^5 = \frac{Ch^5}{32}.$$

Combining the two halves yields a total error of approximately

$$\frac{Ch^5}{16}$$

when the interval is halved. Adaptive integration methods refine the mesh in regions where the integrand is difficult (e.g., rapidly varying or locally unsmooth) and keep larger intervals where the integrand is well-behaved (e.g., smooth and relatively flat).

2 Algorithm Analysis: Adaptive Numerical Integration

General Framework

Branching, bounding, stopping.

Initialization

1. Initialize sum accumulator $X \leftarrow 0$
2. Initialize iteration count $k \leftarrow 0$
3. Initialize good intervals goodIntervals $\leftarrow [\,]$
4. Initialize test intervals testIntervals $\leftarrow [a, b]$
5. Define total tolerance ϵ
6. Define smallest bound $\delta \leftarrow \max(\epsilon/10^k, h_{\min})$

Adaptive Integration Algorithm

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while testIntervals not empty **do**
  (left, right) ← first interval in testIntervals
  mid ← (left + right)/2
  S_1 ← SIMPLE([left, mid])
  S_2 ← SIMPLE([mid, right])
  S ← SIMPLE([left, right])
  $k \leftarrow k + 1$
  $\epsilon_k \leftarrow \epsilon \cdot \frac{\text{right}-\text{left}}{b-a}$
  if $|S_1 + S_2 - S| \leq 15\epsilon_k$ **then**
    Add interval to goodIntervals
    $X \leftarrow X + S_1 + S_2$
    Remove interval from testIntervals
  else
    $h \leftarrow \text{right} - \text{left}$
    if $h > \delta$ **then**
      Replace [left, right] with [left, mid] and [mid, right]
    else
      $X \leftarrow$X + S
        Remove interval (warn: tolerance not met)
      end if
    end if
  end while
  return X

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3 Implementation

4 Testing and Evaluation

4.1 Integration of Smooth Function: $f(x) = \sin(x)$ on $[0, 1]$

Exact Value (J_F): 0.459697694131860
Parameters: $a = 0$, $b = 1$. Adaptive tolerance: $\epsilon = \delta = 10^{-10}$.

Table 1: Results for Smooth Function $f(x) = \sin(x)$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.459862189870785	$10^{-3}(0.164495738924519)$
Simple 3/8 Simpsons Rule		0.459770560550696	$10^{-3}(0.072866418835305)$
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.459697949823821	$10^{-6}(0.255691960326931)$
Composite: Step $N = 10$		0.459697710098337	$10^{-6}(0.015966477251439)$
Composite: Step $N = 20$		0.459697695129542	$10^{-6}(0.000997682048087)$
Composite: Step $N = 50$		0.459697694157399	$10^{-6}(0.000025539070858)$
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 27/53	0.459697694156410	$10^{-10}(0.245496400985701)$
Adaptive 3/8 Simpsons Rule	N_{final}: 20/39	0.459697694173110	$10^{-10}(0.412493372792255)$

4.2 Integration of Rapidly Changing Function: $f(x) = \frac{1}{1+100x}$ on $[10^{-10}, 1]$

Exact Value (J_F): 0.046150205218409
Parameters: $a = 10^{-10}$, $b = 1$. Adaptive tolerance: $\epsilon = \delta = 10^{-10}$.

Table 2: Results for Rapidly Changing Function $f(x) = 1/(1 + 100x)$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.181371867920218	0.135221662701809
Simple 3/8 Simpsons Rule		0.142689160402866	0.096538955184457
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.062756205874418	0.016606000656009
Composite: Step $N = 10$		0.051473363887864	0.005323158669455
Composite: Step $N = 20$		0.047485817370522	0.001335612152113
Composite: Step $N = 50$		0.046278500116435	0.000128294898026
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 189/377	0.046150205252342	$10^{-10}(0.339324124354334)$
Adaptive 3/8 Simpsons Rule	N_{final}: 154/307	0.046150205251867	$10^{-10}(0.334579378091782)$

4.3 Integration of Locally Unsmooth Function: $f(x) = \sqrt{x^2 + A}$

Parameters: $a = -1$, $b = 1$.
The table below tracks the error as parameter A shrinks from 10^{-1} to 10^{-10}.

For $A = 10^{-1}$
Exact Value (J_F): 1.235663960280098

Table 3: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-1}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule		1.120842920135885	0.114821040144213
Simple 3/8 Simpsons Rule		1.213606861689587	0.022057098590511
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		1.235546635525045	0.000117324755053
Composite: Step $N = 10$		1.235663970393828	0.000000010113730
Composite: Step $N = 20$		1.235663943942396	0.000000016337702
Composite: Step $N = 50$		1.235663959860145	0.000000000419953
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 128/255	1.235663960310672	0.000000000030574
Adaptive 3/8 Simpsons Rule	N_{final}: 108/215	1.235663960298045	0.000000000017947

For $A = 10^{-2}$
Exact Value (J_F): 1.034969791615069

Table 4: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-2}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.803325041408059	0.231644750207010
Simple 3/8 Simpsons Rule		1.024509106501572	0.010460685113497
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		1.032386535098423	0.002583256516645
Composite: Step $N = 10$		1.035107454193986	0.000137662578917
Composite: Step $N = 20$		1.034971880027576	0.000002088412507
Composite: Step $N = 50$		1.034969791604493	0.000000000010576
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 160/319	1.034969791645074	0.000000000030005
Adaptive 3/8 Simpsons Rule	N_{final}: 124/247	1.034969791649226	0.000000000034157

For $A = 10^{-3}$
Exact Value (J_F): 1.004647149788814

Table 5: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-3}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.709163618843886	0.295483530944928
Simple 3/8 Simpsons Rule		1.002494897685138	0.002152252103676
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.997049507969993	0.007597641818821
Composite: Step $N = 10$		1.005183195741523	0.000536045952709
Composite: Step $N = 20$		1.004729921988766	0.000082772199952
Composite: Step $N = 50$		1.004648267662287	0.000001117873472
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 150/299	1.004647149817371	0.000000000028557
Adaptive 3/8 Simpsons Rule	N_{final}: 120/239	1.004647149826537	0.000000000037723

For $A = 10^{-4}$
Exact Value (J_F): 1.000579832986624

Table 6: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-4}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.680033332500041	0.320546500486582
Simple 3/8 Simpsons Rule		1.000249948772800	0.000329884213824
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.989528833849176	0.011050999137447
Composite: Step $N = 10$		1.000977378620836	0.000397545634213
Composite: Step $N = 20$		1.000712258259221	0.000132425272598
Composite: Step $N = 50$		1.000596896107151	0.000017063120527
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 128/255	1.000579833023761	0.000000000037137
Adaptive 3/8 Simpsons Rule	N_{final}: 108/215	1.000579833011591	0.000000000024967

For $A = 10^{-5}$
Exact Value (J_F): 1.000069496111630

Table 7: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-5}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.670886370205225	0.329183125906406
Simple 3/8 Simpsons Rule		1.000024999487520	0.000044496624108
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.987529495714036	0.012540000397594
Composite: Step $N = 10$		1.000241923234060	0.000172427122429
Composite: Step $N = 20$		1.000143434237190	0.000073938125565
Composite: Step $N = 50$		1.000089273728380	0.000019777616750
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 106/211	1.000069496149820	0.000000000038198
Adaptive 3/8 Simpsons Rule	N_{final}: 88/175	1.000069496135180	0.000000000023549

For $A = 10^{-6}$
Exact Value (J_F): 1.000008100902585

Table 8: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-6}$

Method	**N / Type**	I_{approx} **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.668000333333250	0.332007767569334
Simple 3/8 Simpsons Rule		1.000002499994870	0.000005600907709
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.986935288883380	0.013072812019205
Composite: Step $N = 10$		1.000069777475790	0.000061676573205
Composite: Step $N = 20$		1.000037137180020	0.000029036277435
Composite: Step $N = 50$		1.000018052683130	0.000009951780554
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 88/175	1.000008100928530	0.000000000025949
Adaptive 3/8 Simpsons Rule	N_{final}: 72/143	1.000008100962260	0.000000000059683

For $A = 10^{-7}$
Exact Value (J_F): 1.000000925219502

Table 9: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-7}$

Method	**N / Type**	I_{approx} **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.667088337021355	0.332912588198147
Simple 3/8 Simpsons Rule		1.000000249999940	0.000000675219553
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.986751189626438	0.013249735593064
Composite: Step $N = 10$		1.000021392935340	0.000020467715848
Composite: Step $N = 20$		1.000010921323710	0.000009996104210
Composite: Step $N = 50$		1.000004688389590	0.000003763170096
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 70/139	1.00000092525413	0.000000000034633
Adaptive 3/8 Simpsons Rule	N_{final}: 58/115	1.00000092523995	0.000000000020452

For $A = 10^{-8}$
Exact Value (J_F): 1.000000104034875

Table 10: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-8}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.666800003333333	0.333200100701542
Simple 3/8 Simpsons Rule		1.000000024999990	0.000000079034876
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.986693352888888	0.013306751145987
Composite: Step $N = 10$		1.000006697775120	0.000006593740253
Composite: Step $N = 20$		1.000003371373280	0.000003267338411
Composite: Step $N = 50$		1.000001380536110	0.000001276501241
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 56/111	1.000000104041540	0.000000000006671
Adaptive 3/8 Simpsons Rule	N_{final}: 48/95	1.000000104038820	0.000000000003953

For $A = 10^{-9}$
Exact Value (J_F): 1.000000011554780

Table 11: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-9}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E\rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.666708830702136	0.333291180852644
Simple 3/8 Simpsons Rule		1.000000002500000	0.000000009054780
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.986675101362649	0.013324910192131
Composite: Step $N = 10$		1.000002111295950	0.000002099741173
Composite: Step $N = 20$		1.000001057896550	0.000001046341770
Composite: Step $N = 50$		1.000000426357300	0.000000414802529
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 46/91	1.000000011555870	0.000000000001094
Adaptive 3/8 Simpsons Rule	N_{final}: 42/83	1.000000011555310	0.000000000000536

For $A = 10^{-10}$
Exact Value (J_F): 1.000000001270607

Table 12: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$, $A = 10^{-10}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule		0.666680000033333	0.333320001237274
Simple 3/8 Simpsons Rule		1.000000000250000	0.000000001020607
Composite Integration (Fixed N and equal spacing)			
Composite: Step $N = 5$		0.986669333528889	0.013330667741718
Composite: Step $N = 10$		1.000000666977750	0.000000665707144
Composite: Step $N = 20$		1.000000333713730	0.000000332443126
Composite: Step $N = 50$		1.000000133805360	0.000000132534755
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: 44/87	1.000000001270740	0.000000000000141
Adaptive 3/8 Simpsons Rule	N_{final}: 38/75	1.000000001270670	0.000000000000072

5 Analysis and Discussion

Adaptive quadrature outperformed all integration techniques for $f(x) = \sin(x)$, $f(x) = \frac{1}{1+100x}$, and $f(x) = \sqrt{x^2 + A}$.

For $f(x) = \sin(x)$, we expect the simple rules to be very accurate very quickly. There are no points where the integrand is poorly behaved, and therefore we can use simple techniques to approximate the integral. Adaptive techniques are indeed more accurate; however, in this case, we must decide if the computational cost is worth the added order of accuracy. Recall that for each iteration, adaptive techniques call three simple Simpson's methods. For $k = 53$, we evaluate 159 simple Simpson's integrations to achieve an accuracy of 10^{-10}. Not included in this study, setting $N = 159$ in composite Simpson's reduces the error to 10^{-12}. Using Simpson's 3/8 rule requires far fewer iterations and should be preferred over the standard Simpson's method.

For $f(x) = \frac{1}{1+100x}$, the adaptive Simpson's technique uses 377 iterations to achieve 10^{-10} accuracy. It is expected that the adaptive technique would outperform other methods, as adaptive methods are designed for these purposes. Not included in the study, setting $N = 377 \times 3 = 1131$ only reduces the error to 10^{-9} for composite Simpson's. Simple Simpson's methods perform adequately because the function remains smooth.

For $f(x) = \sqrt{x^2 + A}$, the simple Simpson's method cannot handle the poorly behaved nature of the function at $x = 0$. This occurs because, in simple Simpson's, a node is placed at 0, causing the nodes to have too much weight in the final approximation. The Simpson's 3/8 rule does not place a node at 0 and, for a single iteration, performs far better. As A gets smaller, the iterations needed for adaptive techniques to reach the desired epsilon also decrease. Not included in the study, setting $N = 87 \times 3 = 261$ causes the error to increase rather than decrease. Setting $N = 100$ does reduce the error, but not to the desired 10^{-10}. In this case, adaptive techniques are clearly the best choice, as they are both more accurate and less computationally expensive.

Overall, we must balance cost with accuracy. For smooth functions, composite integration with a lower N value is preferable. For poorly behaved functions, adaptive techniques are best, as they provide greater accuracy with a similar number of subintervals.